

PRAKHAR SHRIVASTAVA

Mechanical Engineering Undergraduate — Vehicle Dynamics, Suspension & Multi-Body Simulation

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PROFILE

Final-year Mechanical Engineering student specializing in vehicle dynamics and multi-body dynamics (MBD) simulation, with over three years of hands-on MSC Adams modelling of suspension and steering systems — hardpoint definition. Proficiency in complex CAD modelling and detailed engineering drawings using GD&T and bill of materials. Seeking a Vehicle Dynamics Engineer role to build and tune handling, stability, and ride-comfort models for next-generation mobility products.

TECHNICAL SKILLS

- **Vehicle Dynamics & MBD Simulation:** MSC Adams (suspension/steering geometry, hardpoint definition, camber/toe, roll center, Ackermann %, steering effort), ANSYS co-simulation, IPG CarMaker
- **CAD Modelling:** Creo Parametric, SolidWorks, Siemens NX, Solid Edge — part & assembly modelling, GD&T, mass/inertia extraction for simulation import
- **Dynamics & Mechanics:** Handling, stability, and ride comfort analysis; suspension kinematics and compliance, roll centre and load transfer, four-post rig / sinusoidal ride analysis
- **Testing & Validation:** Drop-test and ride-frequency correlation of simulation to hardware, upright load-case extraction, prototype jig/fixture design
- **Scripting & Documentation:** Python, C, MS Office; DFMEA/PFMEA, MOST analysis, BOM management

WORK EXPERIENCE

R&D Intern — Avinya Miltech Pvt. Ltd., Indore (Jun 2026 – Jul 2026)

- Developed a kinematic and dynamic MBD model of a Chemical Mechanical Polishing (CMP) machine in MSC Adams, incorporating contact, friction, RPM, and trajectory optimization.
- Redesigned the 4-bar mechanism of a lens polishing machine to eliminate quick-return motion, applying constraint-based mechanism analysis.
- Performed modal, random vibration, and thermal analyses on an optomechanical assembly in ANSYS; prepared GD&T drawings/BOM in Solid Edge and selected bearings to improve service life.
- Derived conic disc profiles from depth-of-cut vs. radial-distance data; calculated Material Removal Rate (Preston's equation) and built an HTML/Python tool to automate trajectory prediction and re-flattening.

NPD Intern — Gabriel India Ltd., Dewas (May 2025 – Jul 2025)

- Studied twin-tube, mono-tube, and strut damper architectures and mounting systems, building applied knowledge of real-world suspension mechanism design for OEM applications.
- Verified/consolidated BOMs across 1000+ damper assemblies; conducted DFMEA/PFMEA and applied Lean Six Sigma and MOST analysis to streamline assembly-line workflows.

Vehicle Dynamics Member; Multibody Dynamics Head — Team GS Racers, SGSITS Indore (2023 – Present)

- Designed front double-wishbone and rear H-arm suspension in MSC Adams with full hardpoint definition, joint types, and constraint equations; ran MBD simulations for camber/toe gain, roll centre, Ackermann %, and steering effort.
- Performed four-post rig, sinusoidal ride, and drop-test analyses to extract ride frequencies and upright load cases, correlating simulation to hardware, and designed jigs/fixtures for manufacturing.
- Led dynamic MBD analysis of the gearbox to verify torque transfer, backlash, and shaft alignment; simulated differential torque split under varying traction conditions and validated CV joint safety via plunge/articulation angle analysis.
- As Multibody Dynamics Head, led the MBD division across suspension, steering, and powertrain analysis; supervised a cross-functional team and trained juniors on Creo, SolidWorks, and MSC Adams.

TECHNICAL PROJECTS

Reaction Wheel Attitude Control System for Nanosatellites (Jan 2026 – Mar 2026)

- Modelled a reaction wheel assembly (rotor, housing, bearings, motor interface) in Creo Parametric with proper mass balancing; simulated rotational dynamics in MSC Adams (angular momentum transfer, spin-up, attitude response) and built a 3-axis Euler control architecture, validating structural performance in ANSYS.

Rocker-Bogie Suspension for a Planetary Rover (Jan 2026 – Feb 2026)

- Modelled a full rocker-bogie rover (wheel hubs, bogie arms, chassis) in SolidWorks, implemented joints/constraints/contact in MSC Adams, and ran ride analysis over a custom rough-terrain track to evaluate articulation, stability, and load distribution.

ACHIEVEMENTS

- 1st Rank, Validation Event E-BAJA 2026; 7th overall rank, M-BAJA 2024 (also competed in E-BAJA'24/'25).
- AIR 91488 in JEE Mains 2023; School topper, Classes IX–XII.

EDUCATION

Degree	Institute	Year	CGPA
B. Tech, Mechanical Engineering	SGSITS, Indore	2027 (Exp.)	Pending
Class XII (CBSE)	Maharishi Vidya Mandir, Jabalpur	2022	90.0%
Class X (CBSE)	Maharishi Vidya Mandir, Jabalpur	2020	89.4%